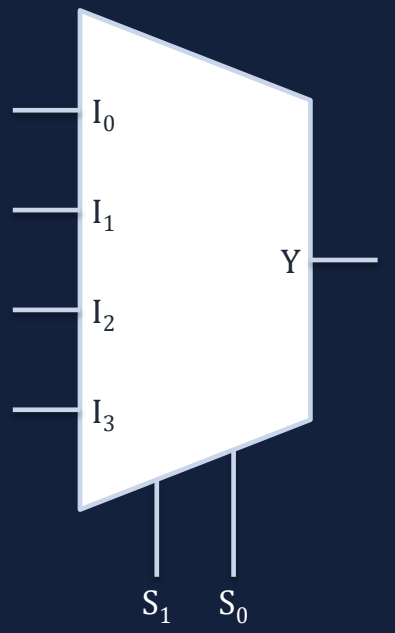
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 13

Multiplexers and Three-State Gates



The 2-to-1 and 4-to-1 multiplexer · functions from a multiplexer · expansion · three-state buffers and buses

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 4, §4.11

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Today’s roadmap

1 §4.11

Multiplexers

Select one of many inputs

2 §4.11

Functions

Any function, no K-map

3 §4.11

Expansion

Bigger MUXes from smaller

4 §4.11

Three-state

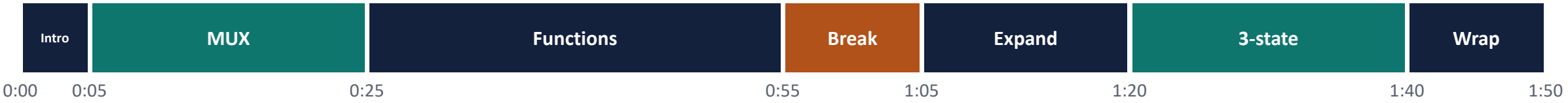
Sharing a bus safely

By the end of class you can…

- Derive a 2-to-1 multiplexer from gates
- Implement an n-variable function on a 2^(n−1)-to-1 multiplexer
- Choose the residual variable to simplify the wiring
- Build larger multiplexers from smaller ones
- Use three-state buffers on a shared bus without contention

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Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	Homework 6 collected
0:05 – 0:25	2-to-1 from gates; the 4-to-1 MUX	Example 1
0:25 – 0:55	Functions on a multiplexer	Examples 2, 3 · Your turn 1
0:55 – 1:05	Break	
1:05 – 1:20	Multiplexer expansion	Example 4
1:20 – 1:40	Three-state buffers and buses	Your turn 2
1:40 – 1:50	Verilog; wrap-up	Lab G preview

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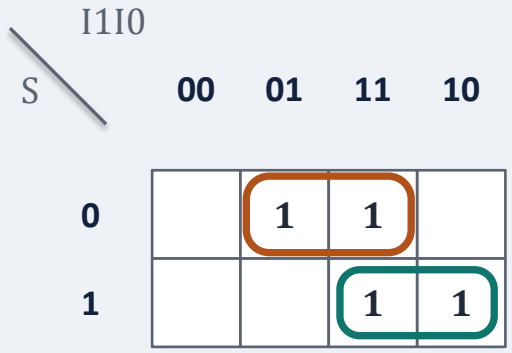
Example 1 — A 2-to-1 MUX from gates

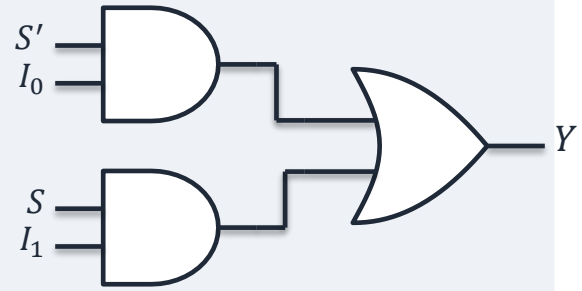
§4.11

EXAMPLE

Output $Y = I_0$ when $S = 0$ and $Y = I_1$ when $S = 1$. Derive the circuit.

S I_1 I_0	Y
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	0
1 1 0	1
1 1 1	1





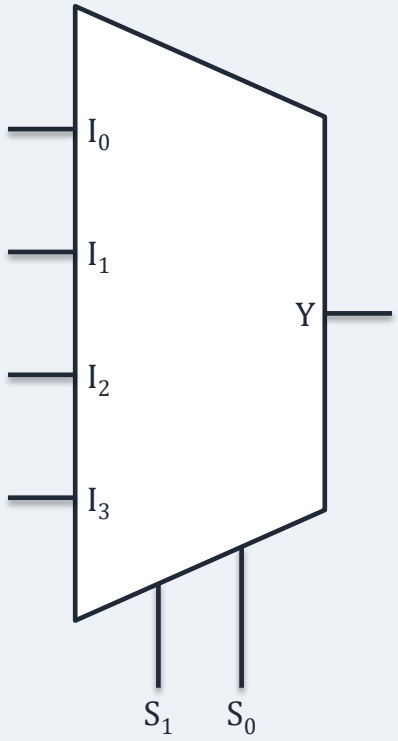
1. Rule: $Y = I_0$ when $S = 0$, $Y = I_1$ when $S = 1$
2. Treat S as one more input: $Y = \Sigma m(1,3,6,7)$
3. Pair m1 (001), m3 (011): $S = 0$, $I_0 = 1 \rightarrow S'I_0$
4. Pair m6 (110), m7 (111): $S = 1$, $I_1 = 1 \rightarrow SI_1$
5. $$Y = S'I_0 + SI_1$$
6. S enables exactly one AND gate

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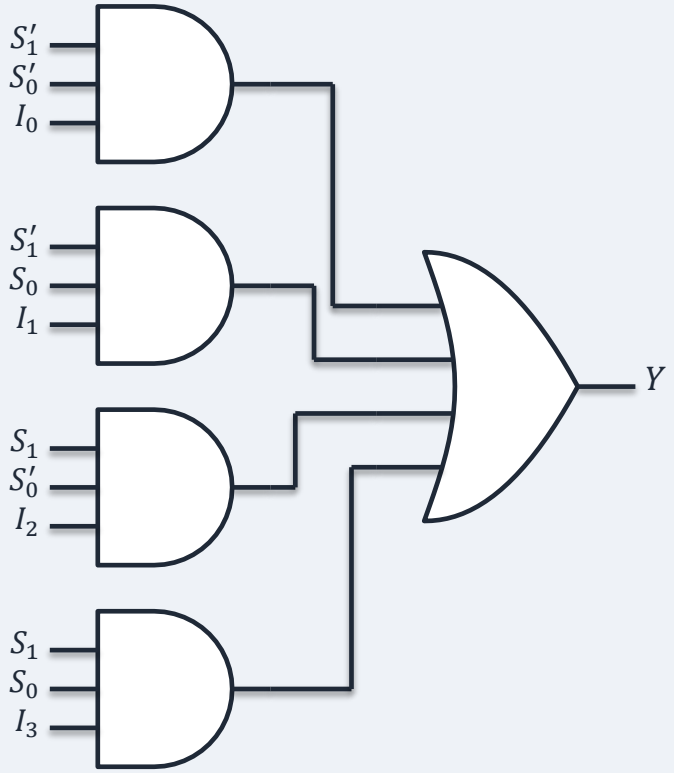
The 4-to-1 multiplexer

§4.11

Symbol



Gate level



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

In general

2^n data inputs, n selects: a decoder whose outputs each gate one input.

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Functions from a multiplexer

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Procedure

An n-variable function fits a 2^{n-1}-to-1 MUX.

Put $n-1$ variables on the select lines; the last one is the residual variable r.

For each select code, look at the two rows $r=0$ and $r=1$.

Wire that data input to 0, 1, r or r'.

F at r = 0	F at r = 1	Data input
0	0	0 (ground)
1	1	1 (V_{CC})
0	1	r
1	0	r'

No K-map is needed: the multiplexer does the decoding. At most one inverter is added.

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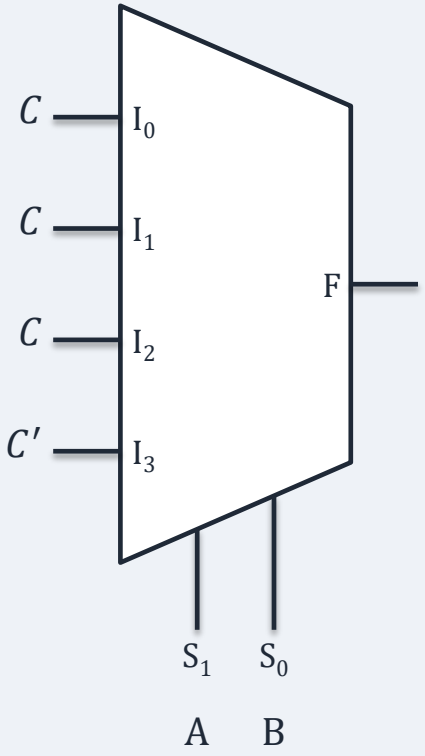
Example 2 — Three variables, 4-to-1

§4.11

EXAMPLE

Implement $F(A, B, C) = \Sigma m(1,3,5,6)$ on a 4-to-1 MUX with A, B on the selects.

A B	C=0	C=1	Input
00	0	1	I0 = C
01	0	1	I1 = C
10	0	1	I2 = C
11	1	0	I3 = C'



1. For each AB, read F at $C = 0$ and $C = 1$
2. $AB = 00$: m0 = 0, m1 = 1 → C
3. $AB = 01$: m2 = 0, m3 = 1 → C
4. $AB = 10$: m4 = 0, m5 = 1 → C
5. $AB = 11$: m6 = 1, m7 = 0 → C'
6. $I_0 = I_1 = I_2 = C, I_3 = C'$
7. Check m6 (110): picks $I_3 = C' = 1$ ✓

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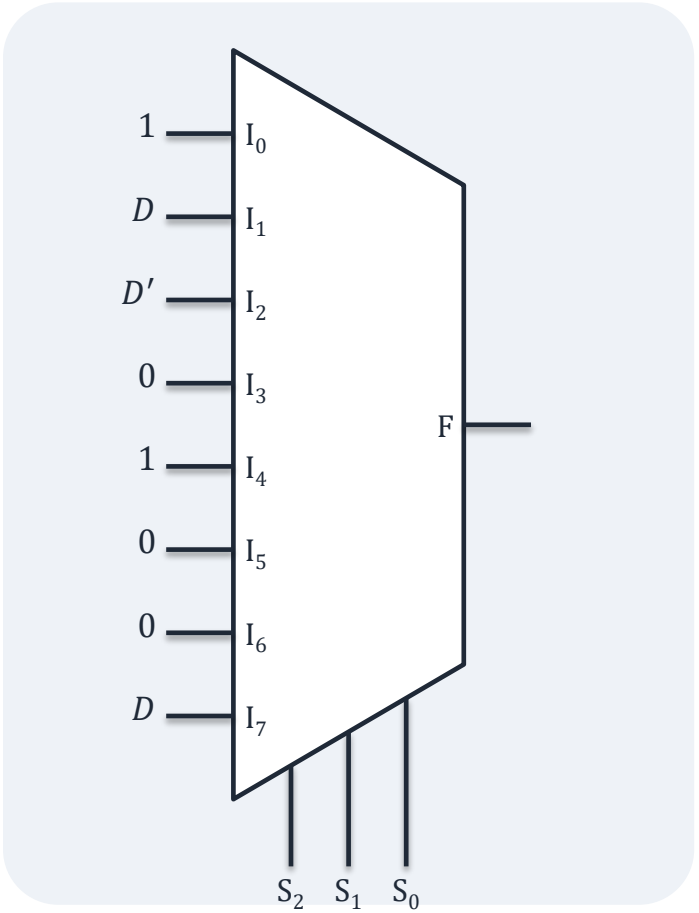
Example 3 — Four variables, 8-to-1

§4.11

EXAMPLE

Implement $F(A,B,C,D) = \Sigma m(0,1,3,4,8,9,15)$ on an 8-to-1 MUX.

A B C	D=0	D=1	Input
000	1	1	I0 = 1
001	0	1	I1 = D
010	1	0	I2 = D'
011	0	0	I3 = 0
100	1	1	I4 = 1
101	0	0	I5 = 0
110	0	0	I6 = 0
111	0	1	I7 = D



1. A, B, C on the selects; D residual
2. Pair values: 00 → 0; 11 → 1; 01 → D; 10 → D'
3. 000: 1, 1 → $I_0 = 1$; 001: 0, 1 → $I_1 = D$
4. 010: 1, 0 → $I_2 = D'$; 011: 0, 0 → $I_3 = 0$
5. 100: 1, 1 → $I_4 = 1$; 101: 0, 0 → $I_5 = 0$
6. 110: 0, 0 → $I_6 = 0$; 111: 0, 1 → $I_7 = D$
7. 1 → V_{CC}, 0 → ground
8. Check m15: 111 → $I_7 = D = 1$ ✓

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Your turn 1 — Choose the residual

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YOUR TURN

≈ 10 min

$F(A, B, C) = \Sigma m(1,2,5,7)$ on a 4-to-1 multiplexer.

(a) Put A, B on the selects (C residual). Find I_0–I_3.

(b) Put B, C on the selects (A residual). Find I_0–I_3.

(c) Which wiring needs no inverter on the residual? Which is cheaper overall?

In (b), pair the minterms by their BC value.

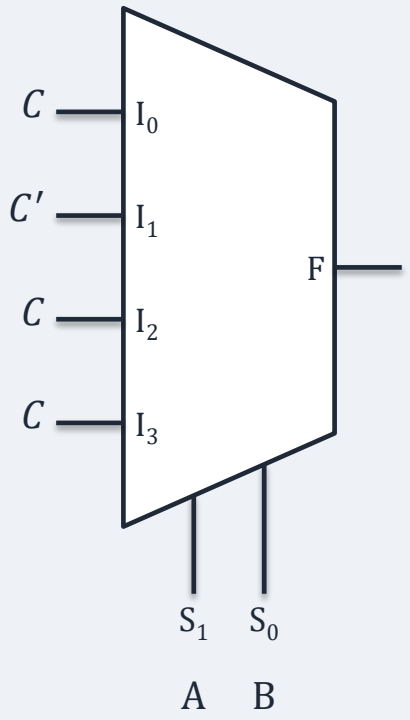
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Your turn 1 — Solution

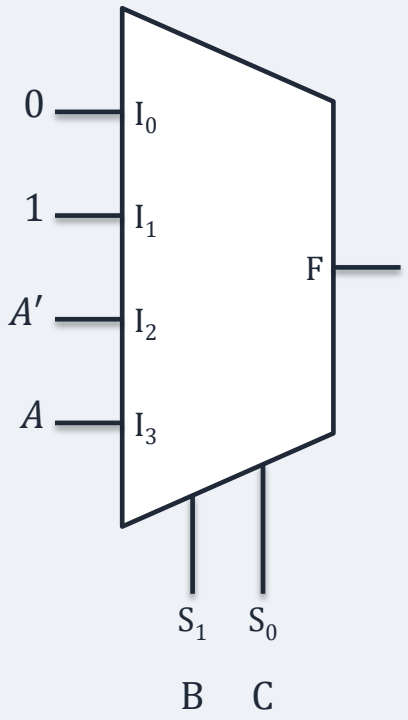
§4.11

SOLUTION

(a) A, B select



(b) B, C select



1. (a) $AB = 00$: m0 = 0, m1 = 1 → C
2. $AB = 01$: m2 = 1, m3 = 0 → C'
3. $AB = 10$: 0, 1 → C; $AB = 11$: 0, 1 → C
4. (b) For each BC: F at $A = 0$, then $A = 1$
5. $BC = 00$: m0, m4 = 0, 0 → 0
6. $BC = 01$: m1, m5 = 1, 1 → 1
7. $BC = 10$: m2, m6 = 1, 0 → A'
8. $BC = 11$: m3, m7 = 0, 1 → A
9. (c) Both need one inverter; (b) needs A on only 2 inputs

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Break

10 minutes — we resume at the 1:05 mark

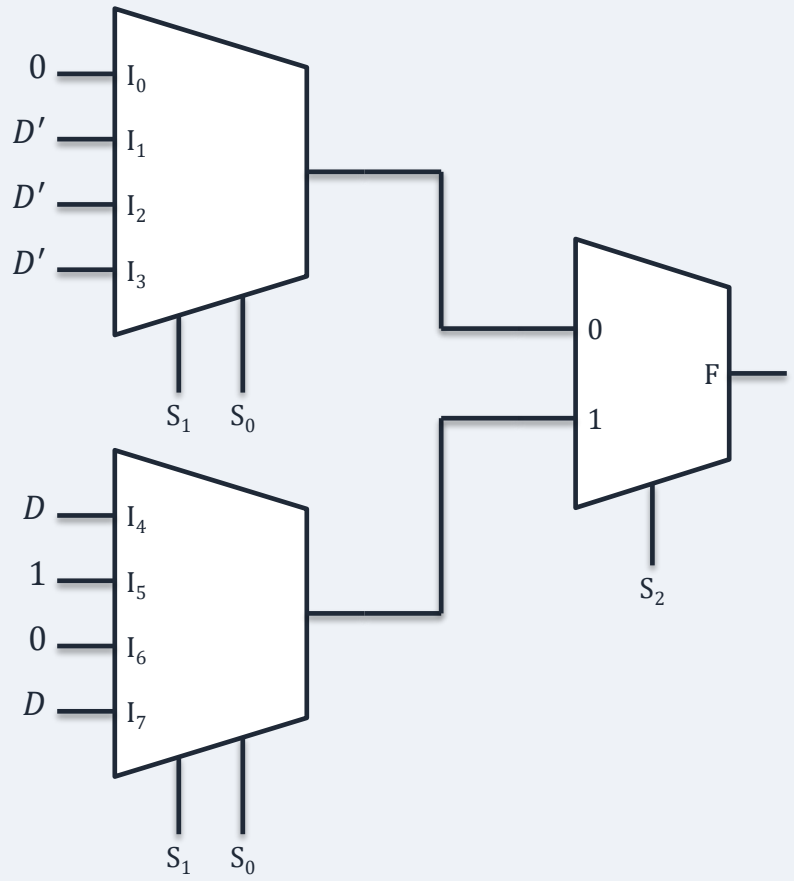
Next: multiplexer expansion and three-state buffers

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Example 4 — Multiplexer expansion

§4.11

EXAMPLE Build an 8-to-1 MUX from 4-to-1 MUXes, then use it for $F = \Sigma m(2,4,6,9,10,11,15)$.



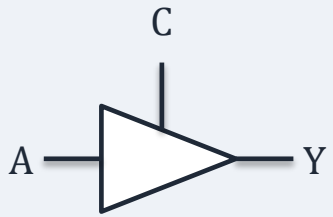
1. S_1S_0 go to both 4-to-1s; the 2-to-1 on S_2 picks one
2. $A, B, C \rightarrow S_2, S_1, S_0$; D residual
3. Pairs ($D = 0$, $D = 1$): 000: 0, 0 → 0; 001: 1, 0 → D'
4. 010: 1, 0 → D'; 011: 1, 0 → D'
5. 100: 0, 1 → D; 101: 1, 1 → 1
6. 110: 0, 0 → 0; 111: 0, 1 → D
7. Check $ABC = 110$: lower MUX, input 2: $I_6 = 0$ ✓
8. Alternative: enables S_2' and S_2, then OR

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Three-state buffers

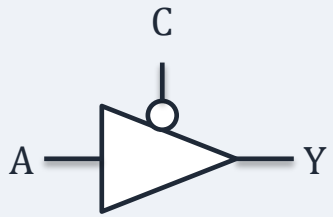
§4.11

Active-high control



C	Y
0	Hi-Z
1	A

Active-low control

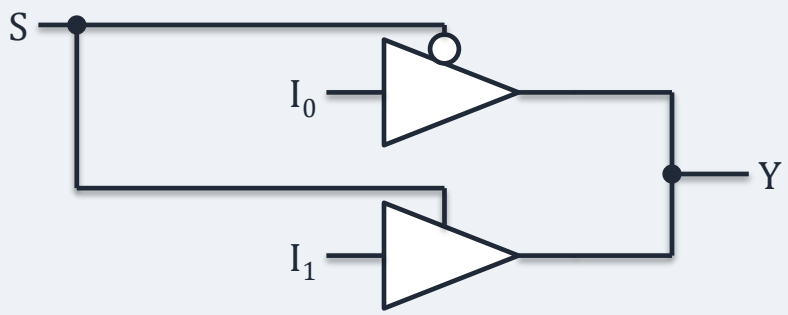


C	Y
1	Hi-Z
0	A

The third state

High impedance (Hi-Z): the output is disconnected, neither 0 nor 1.

Many Hi-Z outputs can share one wire; the one enabled drives it.



A 2-to-1 MUX from two buffers

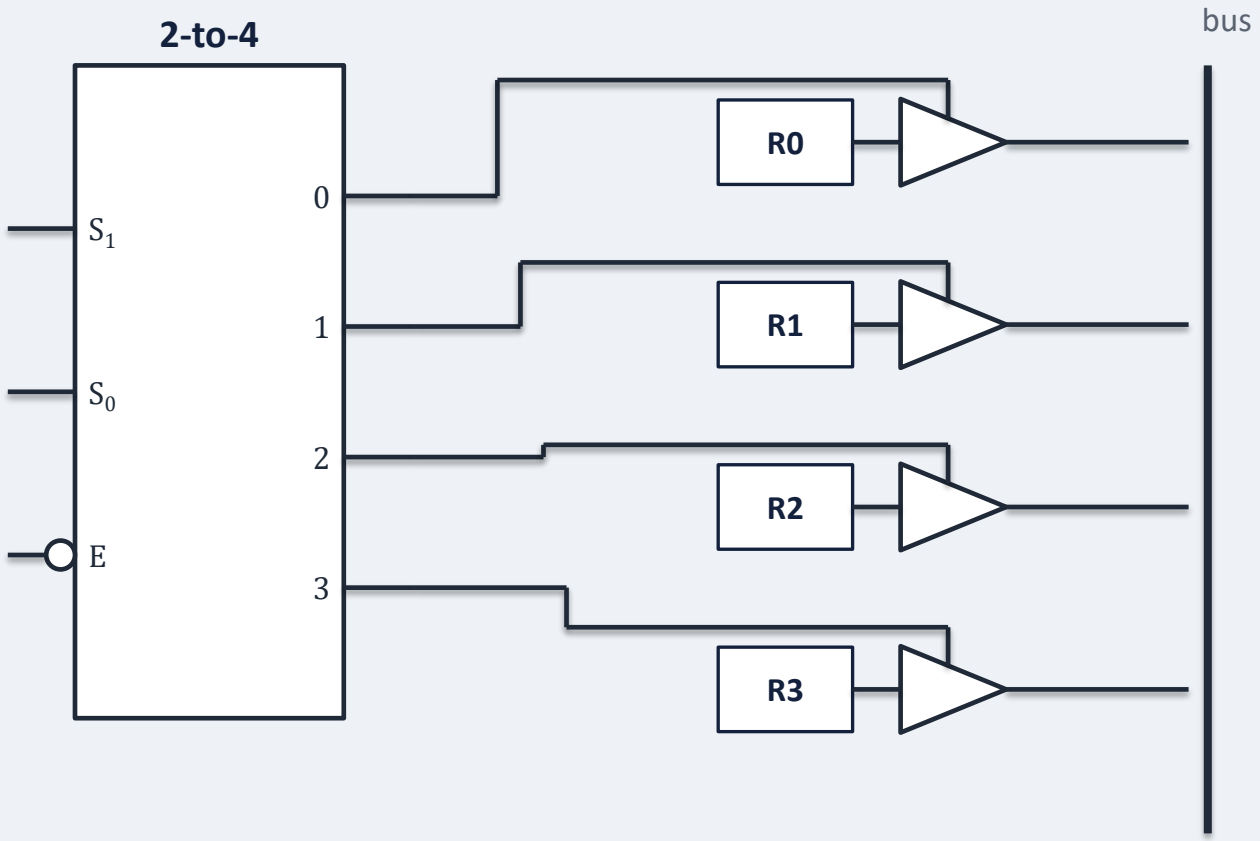
$S = 0$ enables the top buffer, $S = 1$ the bottom one: exactly one drives Y.

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Buses and contention

§4.11

Four registers share one bus line



The rule

Exactly one driver enabled at a time.

A decoder makes that automatic: only one output is ever 1.

$S_1S_0 = 10$: R_2 drives the bus; the others are Hi-Z.

Decoder disabled: nobody drives; the bus is free for another device.

An 8-bit bus uses eight buffers per register, one wire per bit.

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Your turn 2 — Three-state logic

§4.11

YOUR TURN

≈ 10 min

(a) Two enabled three-state outputs drive the same wire, one HIGH and one LOW. What happens?

(b) Build a 4-to-1 multiplexer from three-state buffers and a 2-to-4 decoder. Sketch it.

(c) In the bus of the previous slide, which register drives the bus for $S_1S_0 = 01$? What if the decoder is disabled?

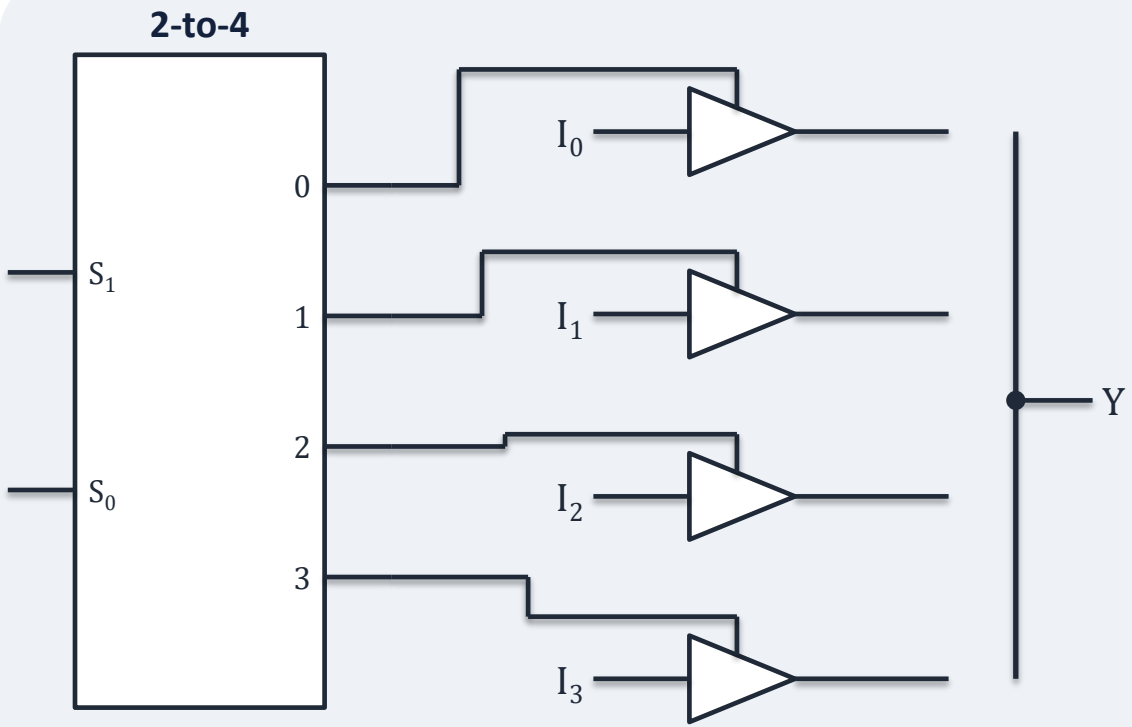
Think of each buffer as a switch.

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Your turn 2 — Solution

§4.11

SOLUTION



1. (a) One output pulls up, the other down: **bus contention**
2. The level is undefined and a large current flows
3. (b) The 2-to-4 decoder turns S_1S_0 into one enable line
4. Output k enables only the buffer carrying I_k
5. Join the four outputs: $Y = I_k$ when $S_1S_0 = k$
6. (c) $S_1S_0 = 01$: decoder output 1 → R_1 drives the bus
7. Decoder disabled: all Hi-Z; the bus floats (free)

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Multiplexers in Verilog

§4.12

Three descriptions

```
// 2-to-1, conditional operator
assign Y = S ? I1 : I0;

// 4-to-1, behavioral
always @(*)
  case (S)
    2'b00: Y = I0;   2'b01: Y = I1;
    2'b10: Y = I2;   default: Y = I3;
  endcase

// three-state buffer
assign Y = En ? A : 1'bz;
```

Notes

`?:` is the Verilog form of $Y = S'I_0 + SI_1$.

In the case form, Y must be declared reg; default avoids an unintended latch.

1'bz is the Hi-Z value: a synthesizer infers a three-state buffer.

Lab G builds the 4-to-1 MUX from gates, then checks it against the case version.

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Concept check

1. How many select lines does a 16-to-1 multiplexer need?

2. What size MUX implements any 5-variable function?

3. What are the four possible values of a data input in the MUX method?

4. Why use a decoder to drive three-state enables on a bus?

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Key takeaways

2-to-1

$$Y = S'I_0 + SI_1$$

2^n-to-1

n select lines

Functions

n vars on 2^{n-1}-to-1

Data inputs

0, 1, r, r'

Three-state

0, 1 or Hi-Z

Bus rule

one driver at a time

A multiplexer is a programmable truth table.

Homework #7 due at the start of Lecture 16

Next: binary adders and subtractors

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